Solar

Boulder Home Pushes for Zero

The Solar Harvest home, built by Boulder-based Ecofutures Building, Incorporated, has been gaining a lot of recognition lately—and for good reason. By achieving 97.7 out of 100 points in an Energy Star audit, it stands as the most energy-efficient house in Colorado. It's the first house in Boulder to pass construction and heating codes without being required to have an electric or gas backup heating system. And it recently won the Colorado Renewable Energy Society's Renewable Energy in Buildings award in the General Housing category.

Built in late fall 2005, the Solar Harvest produces as much energy as it consumes, using only solar energy. Eric Doub, the designer and builder of Solar Harvest as well as the president of Eco-Futures, says that "by meeting this net zero-energy goal, we are striving to build a house that will be affordable and viable 100 years from now, when who knows how much natural gas—and electricity—will cost."

ECOFUTURES BUILDING

The PV system, rated at 6.8 kW, is flush mounted in two arrays on separate portions of the roof.

The spacious two-story home (with 4,585 ft^2 of conditioned space) has five bedrooms and two guest suites, as well

Renewable-Energy Features

- **Solar PV system rated at 6.8 kW.** The PV system is mounted in two arrays on separate portions of the roof. Both arrays are flush mounted on the roof (see photo); one lies at 20° tilt and the other at 40°. The system is larger than the needs of the house require, so it is a net producer of electricity. The local utility, Xcel Energy, purchases the excess generation through a net metering arrangement.

- **Passive-solar design.** The Solar Harvest home has 275 ft^2 of south-facing glass that heats a sunroom. Fans and ducts carry warm air to interior spaces.

- **Solar-thermal heating.** The home has 12 evacuated-tube hot water collectors mounted on the roof, which heat a hot tub, domestic hot water, and living spaces. The collectors are connected to a 6,000-gallon cistern located in the basement, which provides thermal ballast for the home. The hot tub is kept at 103°F. A pump circulates hot water through pipes mounted in the floors for radiant heating in the winter.

- **Natural cooling.** The home has an operable skylight located in a high bay that allows hot air to escape in the summertime. Warm exhaust air also leaves the house through attic vents. Finally, a whole-house fan pulls 1,600 CFM of exhaust air out of the house during the summer nights, which pulls cool, fresh air in from outside.

- **Geothermal and indoor air quality.** Incoming fresh air passes through 260 feet of 6-inch PVC pipe buried 6–8 feet deep surrounding the house for seasonal preheating and precooling. An UltimateAir RecoupAerator energy recovery ventilator (ERV) by Stirling Technologies (Ohio) serves two different purposes, depending on the season. In winter, the ERV recovers about 96% of the heat from outgoing stale air and delivers heated incoming fresh air to all bedrooms and living spaces, bringing the ACH to about 0.3. In summer, the Stirling ERV's EconoCool mode suspends the heat exchange function, and incoming earth-cooled air becomes air conditioning for the home. All bath room exhaust ducts are run through the ERV, while the kitchen is exhausted directly to the outside.

ECOFUTURES BUILDING

The heat from the 275 ft^2 sunroom is carried by fans and ducts to interior spaces.

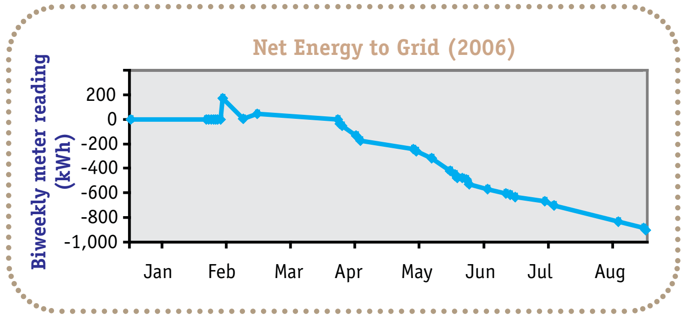

Figure 1. Solar Harvest is tied to the grid and uses a 6.84 kW PV array by Sharp technologies in two mounts connected to two Fronius IG inverters. When the PV system exceeds the home's electricity needs, it sends electricity back to the grid.

as a 400 ft^2 unfinished garage and a 600 ft^2 unfinished carport. The wood-framed home relies on a mix of energy efficiency, renewable technologies, and conservation measures to secure zero-energy status. Solar Harvest is tied to the grid and uses a 6.84 kW PV array by Sharp technologies in two mounts connected to two Fronius IG inverters. The panels provide enough electricity to more than meet the home's energy needs (see Figure 1).

The building envelope is made up of FSC-certified 2 x 6 studs with 1-inch resilient channel at 24 inches OC. Double 5/8-inch Sheetrock for distributed thermal mass is also utilized. Seven inches of the spray foam insulation is used for the walls, and 12 inches of Icynene sprayfoam insulation is used for the ceilings. Above-ground walls rate R-34 and the ceiling rates R-45. One-inch rigid extruded polystyrene foam (XPS) is encased with 1/8-inch GrailCoat waterproofing exterior, and walls are slightly thicker than the standard 2 x 6 wall.

A 6,000-gallon superinsulated tank designed by Doub stores hot water at 170°F–190°F. The hot water comes courtesy of 12 salvaged Novan solar-thermal flat-plate collectors set up in a drainback system, which acts as a thermal battery. The tank is constructed with Greenblock insulated concrete form (ICF) walls, Icynene insulation, and recycled polyiso foam. Coils of copper tubing act as heat exchangers, transferring heat to domestic hot water, the staple-up radiant floor space-heating system, and the 360-gallon outdoor hot tub, which is kept at 103°F. The water tank can heat the house to 68°F for eight consecutive days.

The house utilizes geothermal preheating and precooling for incoming fresh air. The geothermal system consists of 260 feet of 6-inch PVC pipe buried 6–8 feet below ground. With this system, the energy recovery ventilator (ERV) is able to run continuously even if exterior temperatures are below 28°F, and can achieve a maximum heat recovery of 95%. In the winter of 2005-2006, when the exterior temperature was 4°F, incoming fresh air to the ERV was measured at 38°F. And in summer, 95°F is easily converted into 60°F at 200 CFM. An UltimateAir RecoupAerator ERV functioning at 96% heat recovery efficiency ensures continuous fresh-air exchange into the airtight building with 0.1 natural ACH or lower.

The Solar Harvest has average Xcel Energy bills of $12 per month for 2 therms of natural gas used for the range and dryer, and $6 per month connectivity fee for less than zero electricity usage (reconciled annually, the Doub family receives a small check from Xcel, about enough to go out to dinner). The extra electricity generated by the home more than offsets the energy in 2 therms of gas, so that in all, the home has a net negative energy bill for the year.

The house was part of the Boulder Tour of Solar and Green-Built Homes in fall 2005. Doub says that he is "living in the most comfortable home I have ever experienced. In this climate and with this high-performing envelope, when the sun is shining on a winter day, it almost feels like cheating. It just seems almost too easy now, to be better than net zero-energy."

Doub says that the next challenge is trying to green their car, pointing out that an average Boulder household's greenhouse gas impact is 42% house and 58% transportation. Currently, Doub wants to use the outlets in the carport to plug in the family's Prius. He says that the PV system's extra 10–15 kWh per day should help transport them locally. "We've built a home that gets an infinite number of miles per gallon, so to speak, and we want to be driving cars we can feel good about, too. Over 1,200 people have toured Solar Harvest since October 2005, and I think I am as energized by sharing our success as visitors are inspired. It's the right time in history to have designed and built a home that has a net negative energy bill for the year; people are captivated, fascinated—and even though I live here, so am I!"

—Elka Karl

*Elka Karl is an associate editor with **Home Energy** magazine.*